Fix First and Second Class's broken delivery promises first, then reduce Standard Class's late orders, the largest single tier by dollars.

1

First and Second Class

Take twice as long as promised

2

Standard Class

Largest single tier by dollars

For: a supply chain / logistics operations manager at DataCo.

What we assumed, and what we worked out from the data.

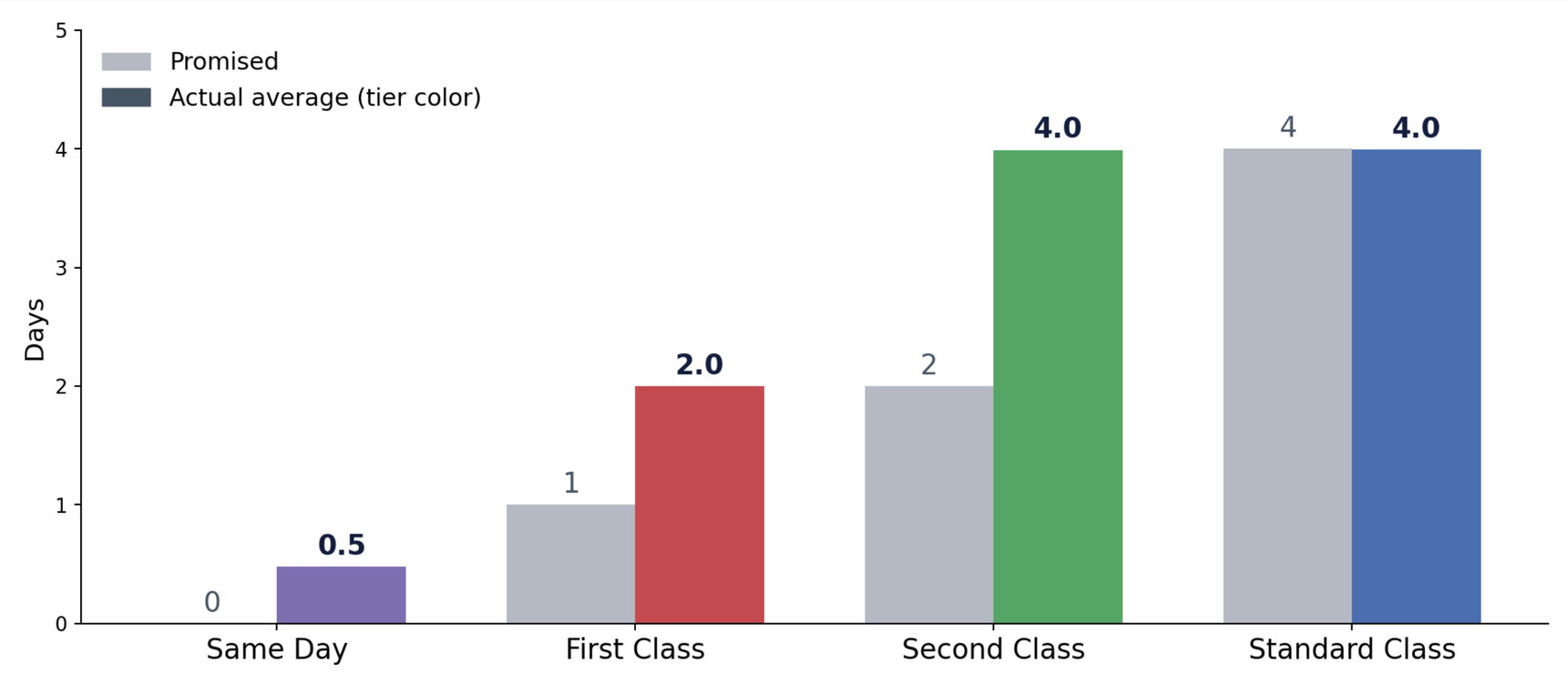
Assumed

- Day counts were recorded correctly
- $15 / $20 / $40 handling cost per late order (outside sources)
- Sales on late orders = revenue exposed
- 80/20 split, seed 42, model settings were my choices
- The data reflects real DataCo operations

Worked out from the data

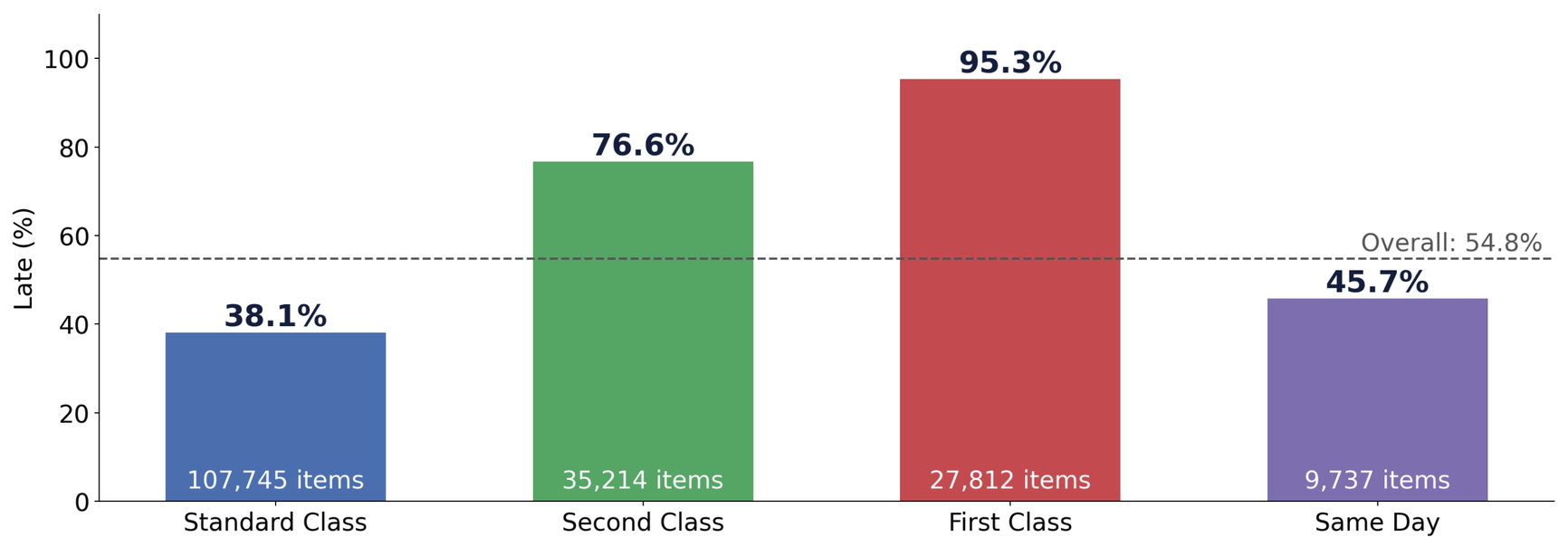
- Late = actual days > scheduled days
- Promised days: 0 / 1 / 2 / 4 by class
- Canceled and fraud orders are never flagged late
- Each row is an order item: 180,508 items, 65,741 orders
- First Class takes exactly 2 days on every order

First and Second Class promise 1 and 2 days, and take about twice as long.



FOUND IN DATA Promised days (0 / 1 / 2 / 4) are read from the scheduled-days column. First Class takes exactly 2 days on every order.

More than half of orders arrive late, and the rate ranges from 38% to 95% by shipping class.

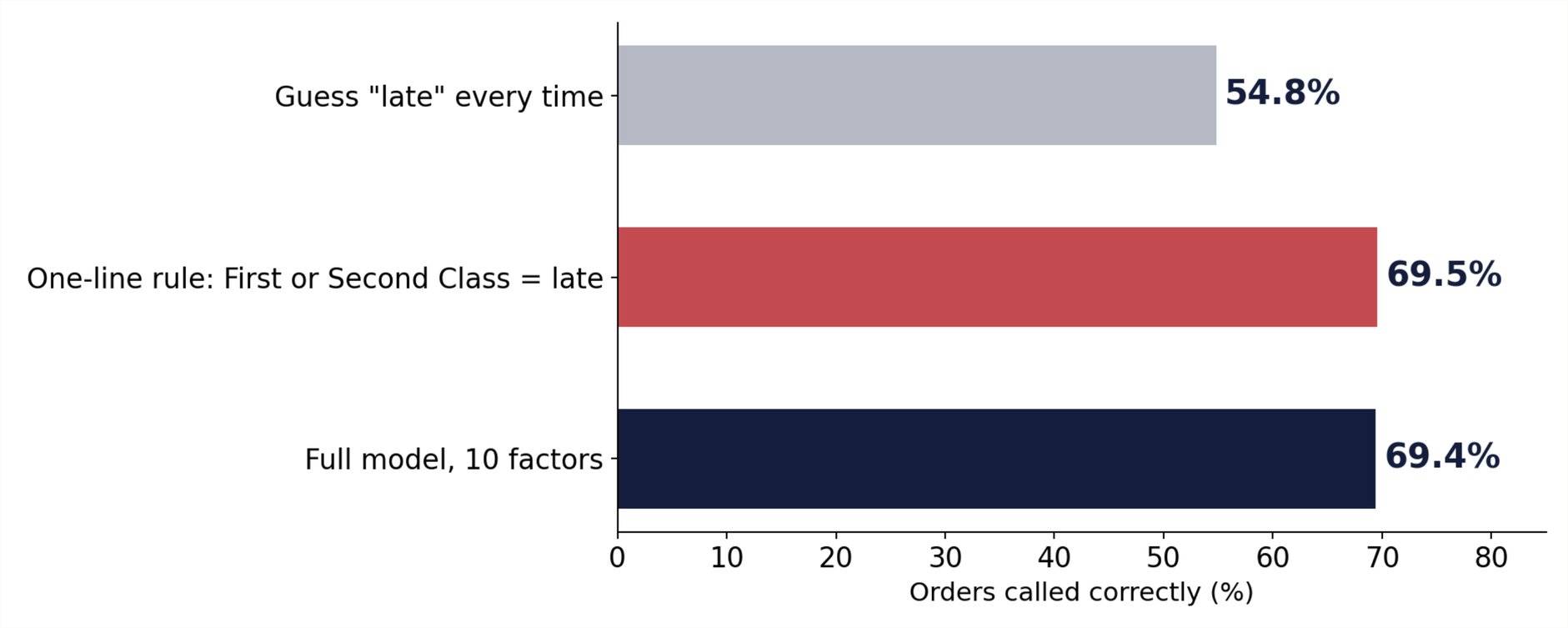


FOUND IN DATA Late = actual days exceeded scheduled days. The description file gives no formula; Delivery Status agrees, except canceled and fraud orders.

ASSUMED The day counts were recorded correctly.

FOUND IN DATA Rates include 7,754 canceled and suspected-fraud items, which are always flagged 0% late. Without them the range is about 40% (Standard Class) to 100% (First Class).

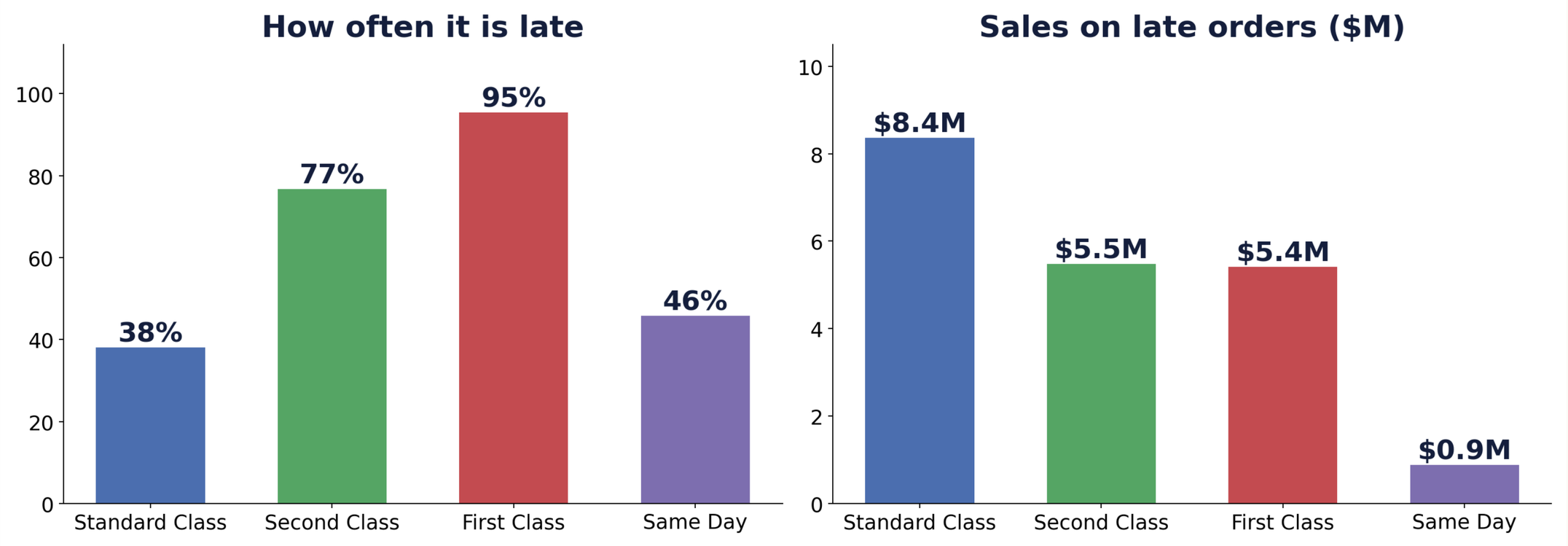
A one-line rule matches a full 10-factor model on accuracy.



FOUND IN DATA The rule matches the full model on accuracy, so class carries the signal. Recall and AUC: on the test set the rule catches 54% of late items (precision 85%); logistic regression catches 54% and random forest 58%, with AUC 0.74 for both models. AUC is reported for the models only, since the rule gives a yes/no call.

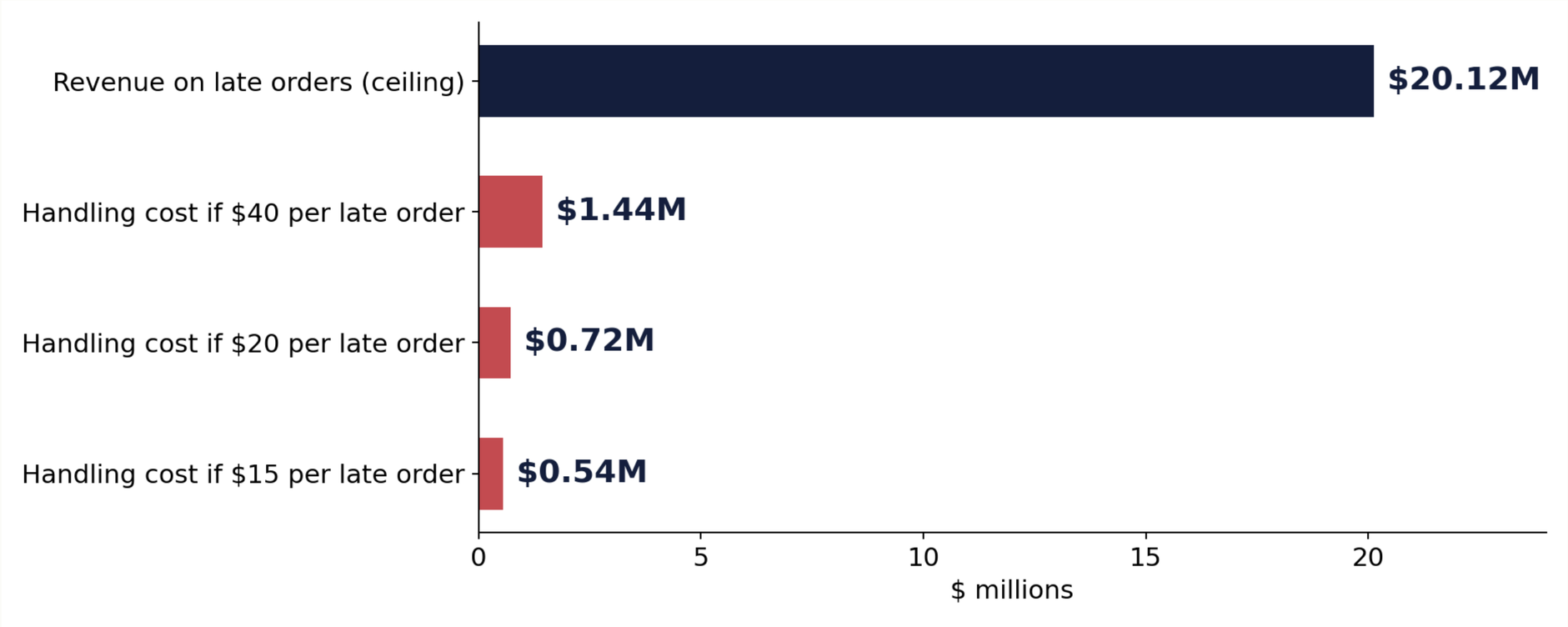
ASSUMED 80/20 split and model settings were my choices; they held up in a tuning check.

Standard Class is the safest tier and the largest single tier to get wrong, and those are not the same problem.



ASSUMED Revenue at risk = sales on late orders. It is exposure, not confirmed loss.

The ceiling is $20.1M; the cost of handling late orders is an estimated $0.5M to $1.4M.



ASSUMED $15 / $20 / $40 per late order is from outside industry sources, not DataCo data.

FOUND IN DATA Canceled and fraud orders are never marked late, so lost revenue cannot be measured here.

We recommend fixing the First and Second Class promises first, then Standard Class's late orders.

1

First Class

Find where the extra day is lost, and keep the 1-day promise

2

Second Class

Same check: promised 2 days, takes about 4

3

Standard Class

Cut late orders (largest single tier): 14,993 late orders, $8.4M exposed

FOUND IN DATA First and Second Class together hold more late revenue ($10.9M) than Standard Class ($8.4M).

ASSUMED The 1-day and 2-day promises stay as they are. We fix delivery, not the commitment.

What this data cannot tell us.

- Actual loss: canceled and fraud orders are never marked late, so the data cannot show whether lateness causes cancellation.
- Why First Class takes 2 days: the data shows that it does, not why.
- "Late" is built from days, so class and lateness are tied by definition. The data may be rule-generated.
- Late items earn $0.78 less profit on average, but the gap is not statistically significant (p = 0.11).

FOUND IN DATA Late items earn $0.78 less profit on average ($21.62 vs $22.40); not significant, p = 0.11.

Next step: find where First Class loses its extra day, then test the fix on one group.

1

Trace the extra day

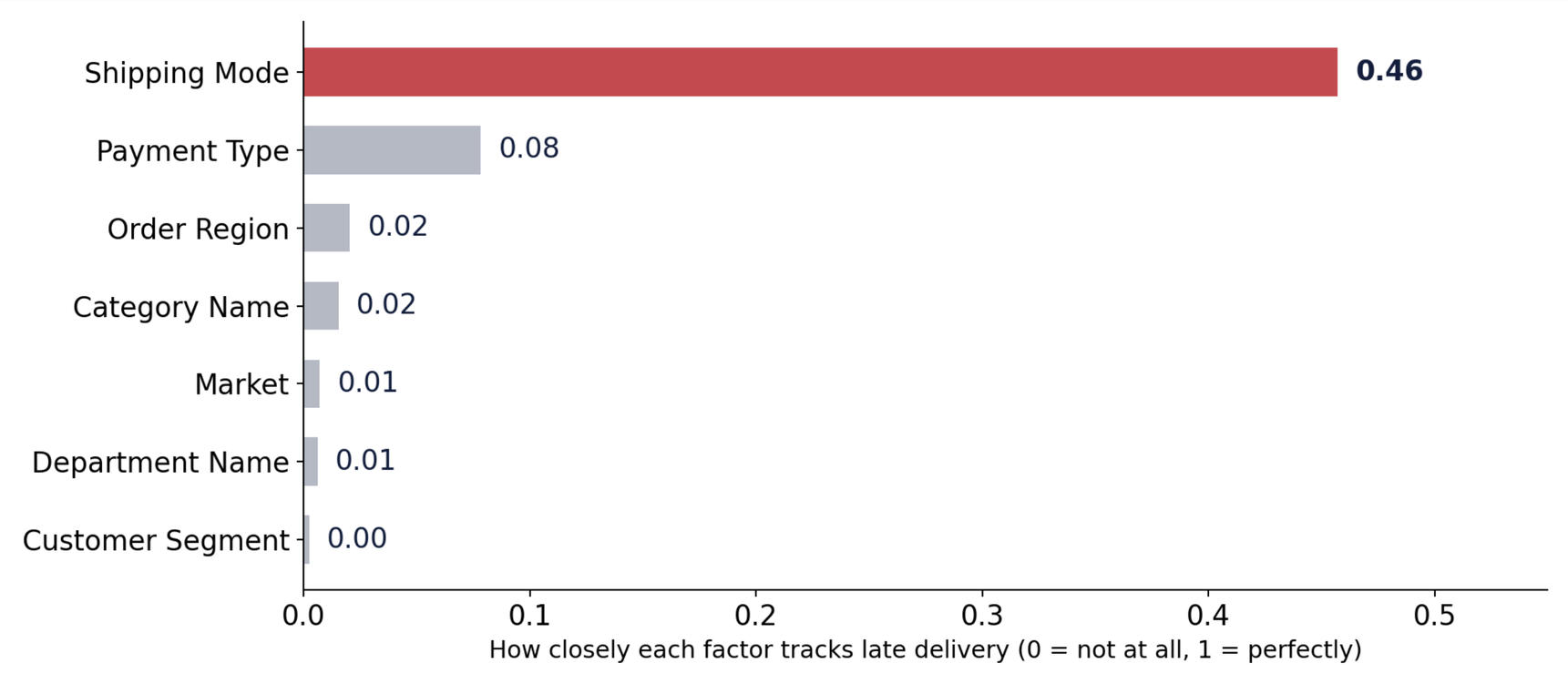
Where do First Class orders lose their day?

2

Test the fix

Apply it to one group, compare late rates

Appendix: Shipping Mode is the only factor with a strong link to late delivery.



FOUND IN DATA Strength of link measured with Cramér's V (0 to 1).

Appendix: how we know an order is late.

- All 98,972 late order items took more days than scheduled (1 to 4 over). No on-time order did.
- Delivery Status agrees: every "Late delivery" order is flagged late; "Shipping on time" is exactly the scheduled days; "Advance shipping" is 1 to 2 days early.
- 4,423 order items took longer than scheduled but are not flagged. All are canceled or suspected fraud.
- The description file gives no formula, so this rule is deduced.

FOUND IN DATA Rule deduced from the day counts and cross-checked against Delivery Status.

Appendix: the numbers we chose and where they come from.

- Handling cost: $15, $20, $40 per late order on 36,043 late orders gives $0.54M, $0.72M, $1.44M. From outside industry sources (HubBox, Locus.sh, OneRail and similar), not DataCo.
- Loss scenarios: if 1%, 5%, 10% of late sales were lost, $0.20M, $1.01M, $2.01M. Illustrative, not findings.
- Cleaning: dropped Product Description (100% missing), Order Zipcode (86% missing), and rows missing Customer Lname or Customer Zipcode (11 rows). Each row is an order item (180,508 items in 65,741 orders), so counts of late orders are by unique order. 2018 is a partial year.
- Model: 80/20 stratified split, seed 42. Scheduled days left out because they overlap with Shipping Mode.

ASSUMED These are my choices or outside estimates; the data cannot confirm them.

Appendix: the model settings held up when I tested alternatives.

- Logistic regression: regularization strength from 0.01 to 10 gave the same score (AUC 0.741).
- Random forest: depth 10 and 100 trees were my starting choices. Depth 5, 10, 15 scored 0.740, 0.742, 0.740; on the order-grouped check depth 15 is lower (0.732). I also checked 200 trees, with no change.
- On the held-out test set: logistic regression 69.4% accuracy, random forest 69.2%. Splitting by order instead of by row gave 69.9% and 69.5%, so the row split did not inflate them.
- Decision cutoff: 0.5 gives 69.4% and 0.6 gives 69.5%; lower cutoffs score worse. I kept 0.5: it is the default, within about 0.1 point of the best cutoff, and I did not weigh the cost of a missed late order against a false alarm.
- Different split seeds move accuracy by about 0.6 points, so the rule and the models are about equal.

ASSUMED Settings were searched on the training split only. The test set was used to report results, not to pick settings.